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(54) **BARRIER STRUCTURE CONFIGURED TO INCREASE PERFORMANCE OF III-V DEVICES**

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(58) **Field of Classification Search**
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(56) **References Cited**
U.S. PATENT DOCUMENTS

2010/0219452 A1 9/2010 Brierley
2012/0056191 A1 3/2012 Endo et al.
(Continued)

OTHER PUBLICATIONS

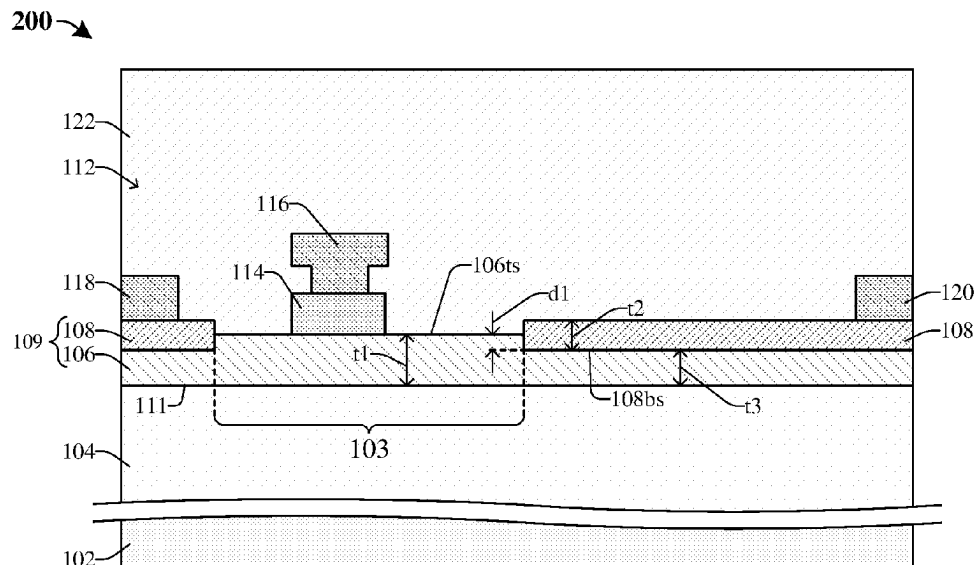
Meneghini et al. "Technology and Reliability of Normally-Off GaN HEMTs with p-Type Gate." *Energies* 2017, 10, 153; doi:10.3390/en10020153, published Jan. 25, 2017.
(Continued)

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(57) **ABSTRACT**

Various embodiments of the present disclosure are directed toward an integrated chip including an undoped layer overlying a substrate. A first barrier layer overlies the undoped layer. A doped layer overlies the first barrier layer. Further, a second barrier layer overlies the first barrier layer, where the second barrier layer is laterally offset from a perimeter of the doped layer by a non-zero distance. The first and second barrier layers comprise a same III-V semiconductor material. A first atomic percentage of a first element within the first barrier layer is less than a second atomic percentage of the first element within the second barrier layer.

20 Claims, 6 Drawing Sheets



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(56) **References Cited**

U.S. PATENT DOCUMENTS

2017/0278960	A1	9/2017	Shiue et al.
2018/0026106	A1 *	1/2018	Yeh H01L 29/205 257/76
2020/0127116	A1	4/2020	Chen et al.
2021/0320199	A1	10/2021	Liu et al.

OTHER PUBLICATIONS

Notice of Allowance dated Mar. 20, 2023 for U.S. Appl. No. 16/872,551.

Non-Final Office Action dated Aug. 30, 2022 for U.S. Appl. No. 16/872,551.

Eastman, L. "Undoped AlGaIn/GaN HEMTs for Microwave Power Amplification" IEEE Trans. on Elec. Dev. Vol. 48, No. 3 Mar. 2001 pp. 479-485 (Year: 2001).

Anderson, T. "An AlN/Ultrathin AlGaIn/GaN HEMT Structure for Enhancement-Mode Operation Using Selective Etching" IEEE Elec. Dev. Lett. Vol. 30 No. 12 Dec. 2009 pp. 1251-1253 (Year: 2009).

* cited by examiner

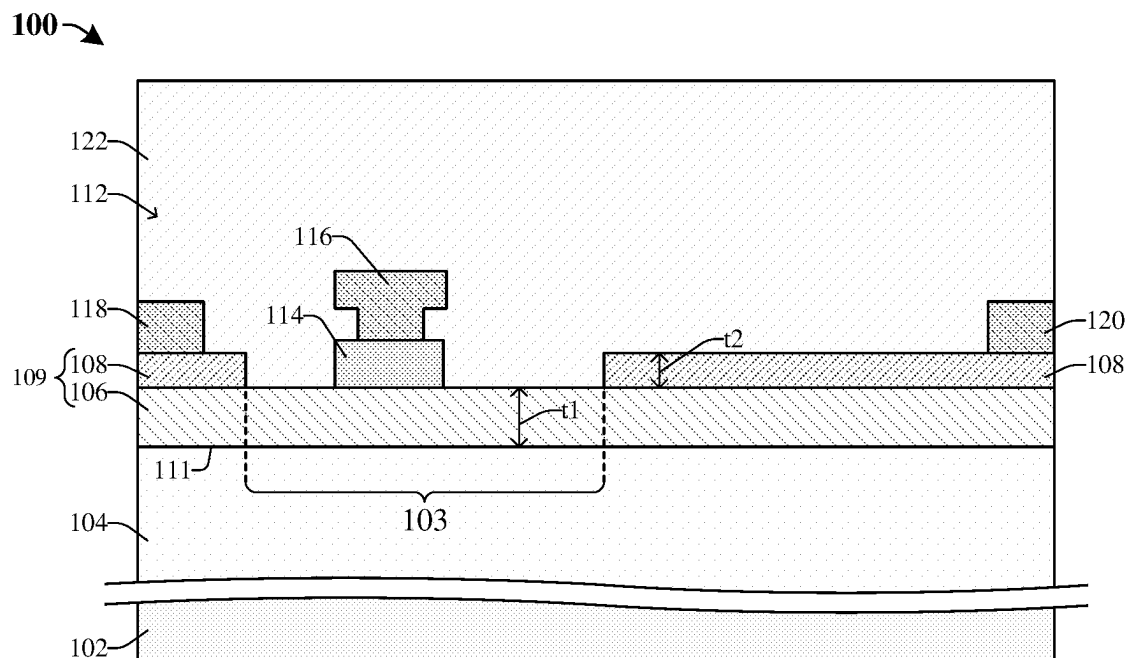


Fig. 1

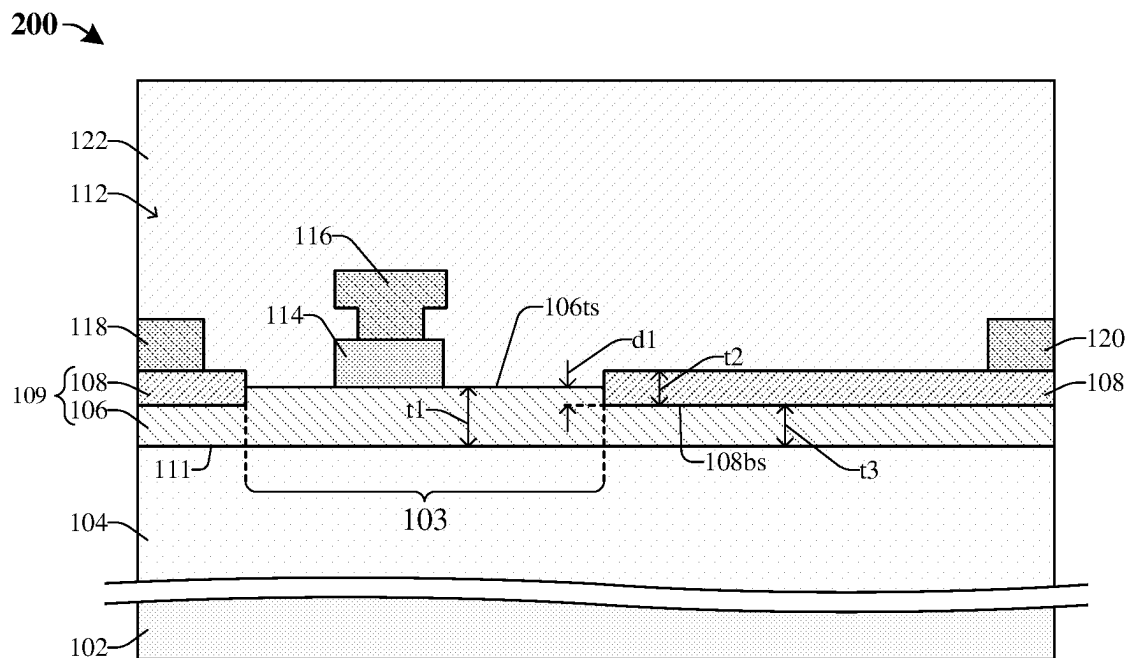


Fig. 2

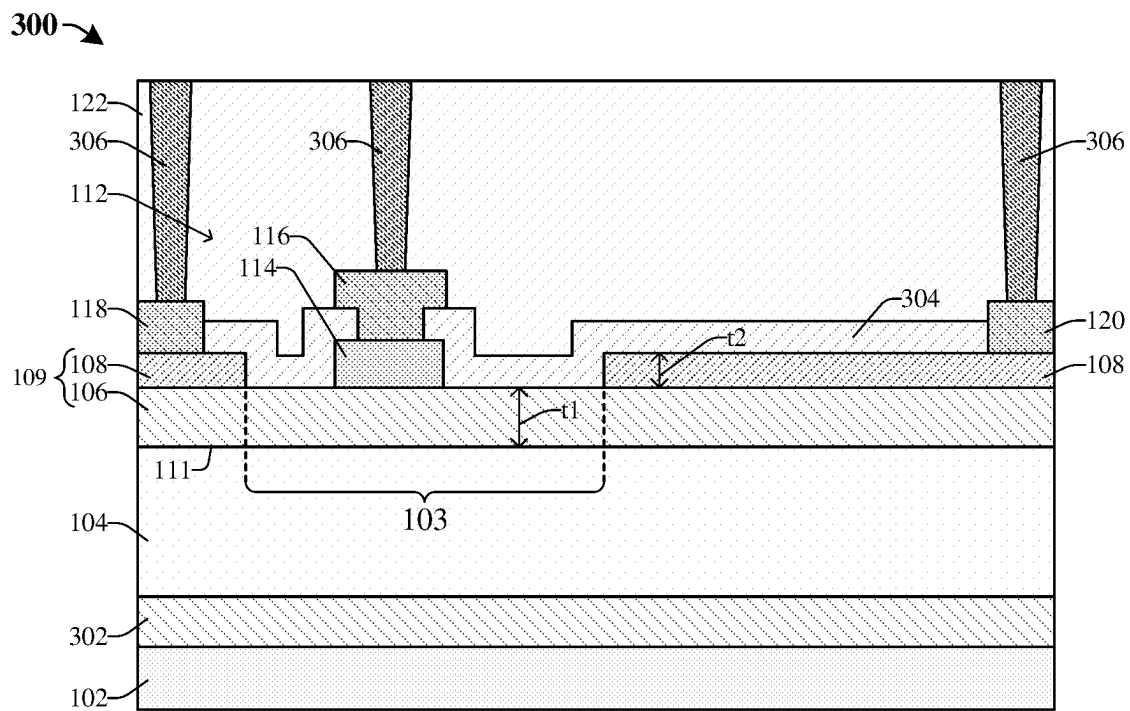


Fig. 3

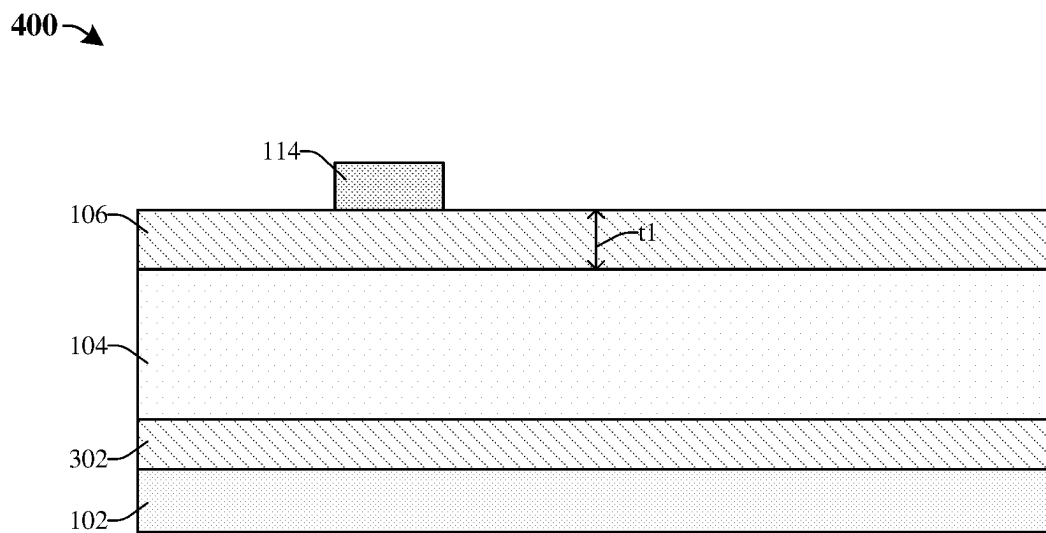


Fig. 4

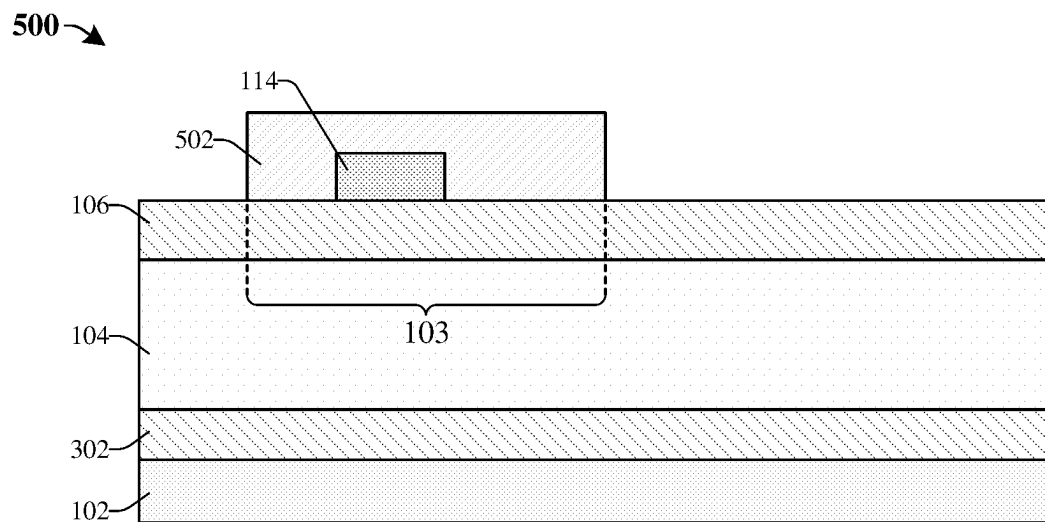


Fig. 5

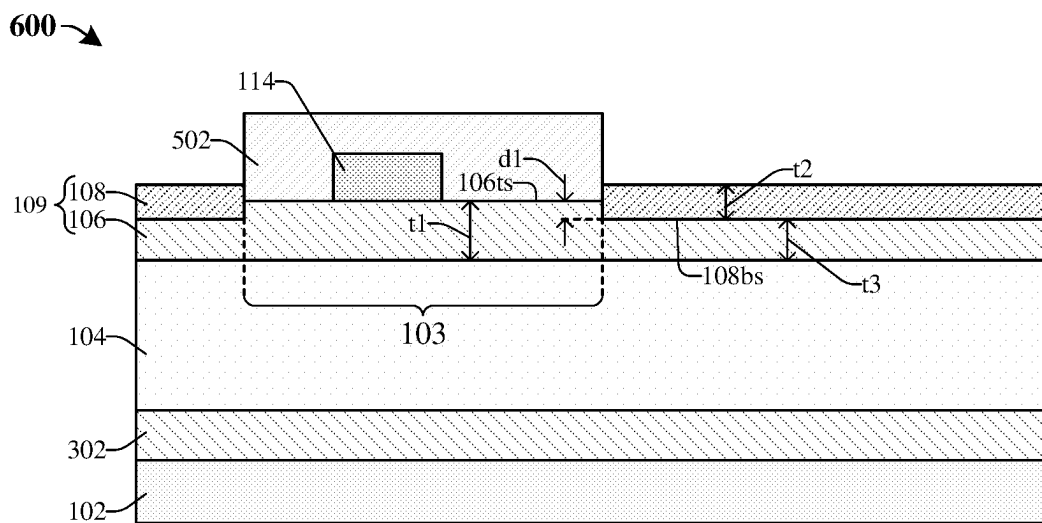


Fig. 6

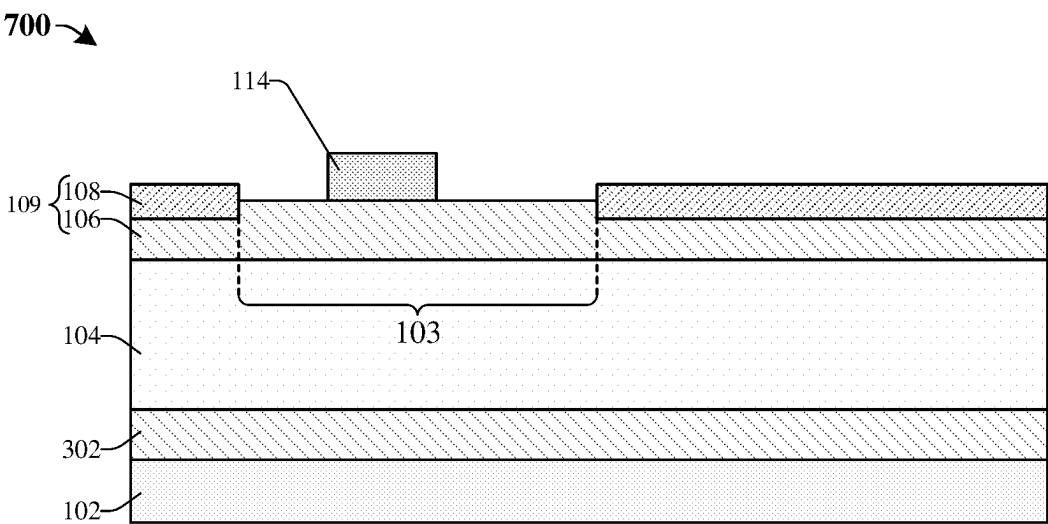


Fig. 7

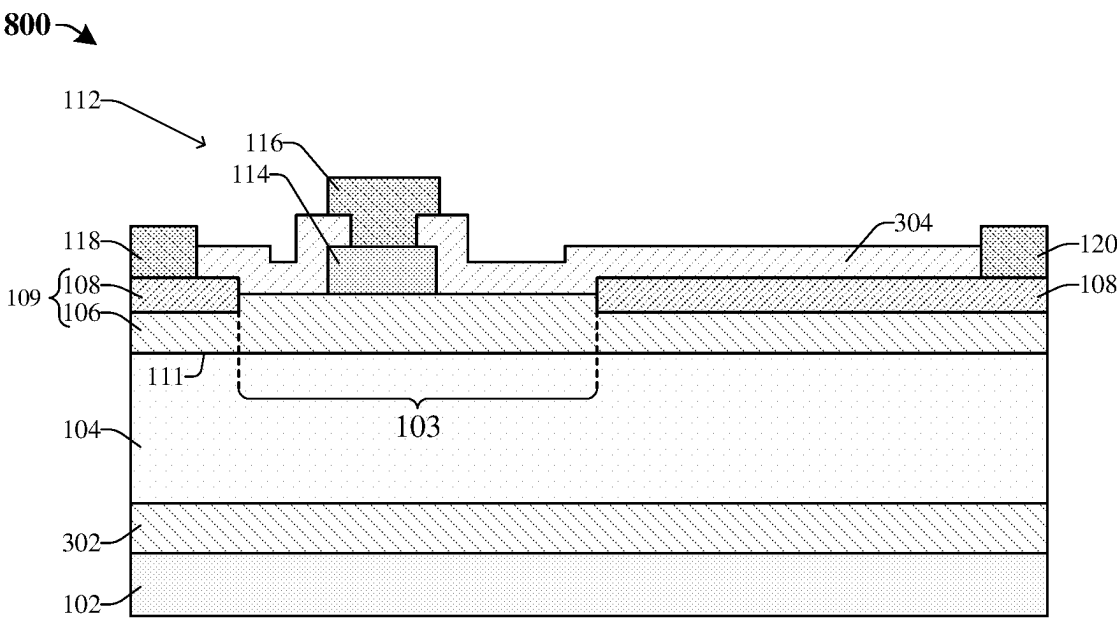


Fig. 8

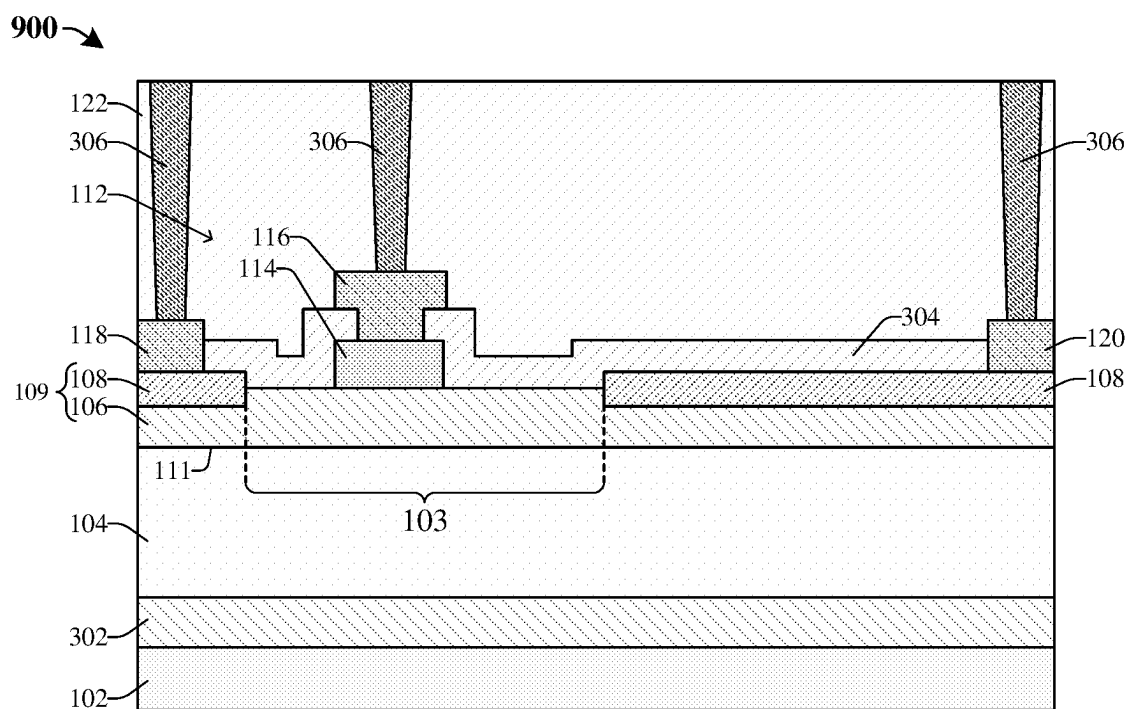
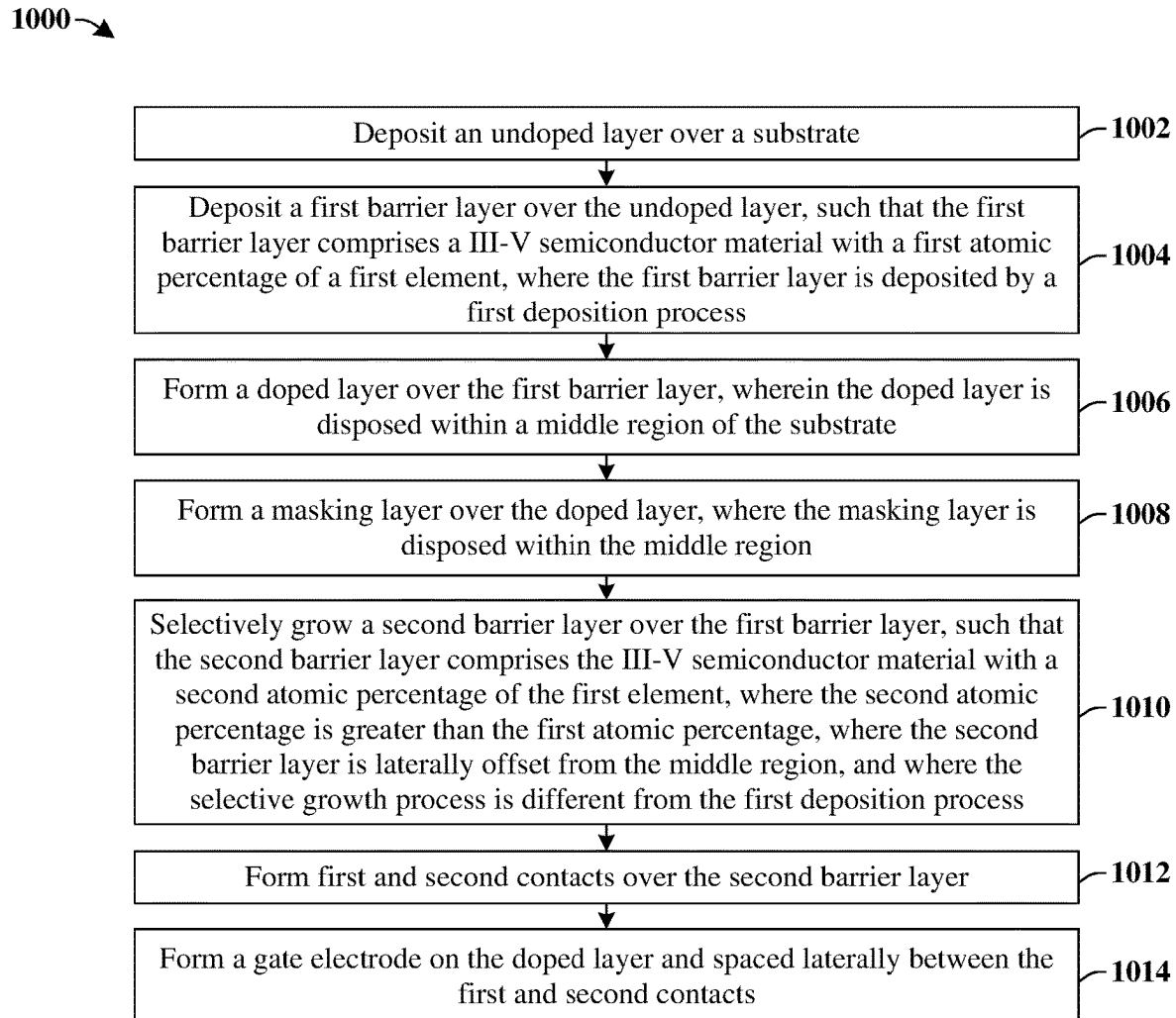


Fig. 9

**Fig. 10**

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BARRIER STRUCTURE CONFIGURED TO INCREASE PERFORMANCE OF III-V DEVICES

REFERENCE TO RELATED APPLICATIONS

This application is a Divisional of U.S. application Ser. No. 16/872,551, filed on May 12, 2020, which claims the benefit of U.S. Provisional Application No. 62/968,212, filed on Jan. 31, 2020. The contents of the above-referenced patent applications are hereby incorporated by reference in their entirety.

BACKGROUND

Modern day integrated chips comprise millions or billions of semiconductor devices formed on a semiconductor substrate (e.g., silicon). Integrated chips (ICs) may use many different types of transistor devices, depending on an application of an IC. In recent years, the increasing market for cellular and radio frequency (RF) devices has resulted in a significant increase in the use of high voltage transistor devices. Thus, high electron mobility transistor (HEMT) devices have been receiving increased attention due to high electron mobility and wide band gaps compared to silicon-based semiconductor devices. Such high electron mobility and wide band gaps allow improved performance (e.g., fast switching speeds, low noise) and high temperature applications.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates a cross-sectional view of some embodiments of an integrated chip including a high electron mobility transistor (HEMT) device having a first barrier layer underlying a second barrier layer.

FIGS. 2 and 3 illustrate cross-sectional views of some alternative embodiments of the integrated chip of FIG. 1.

FIGS. 4-9 illustrate cross-sectional views of some embodiments of a method of forming an integrated chip with an HEMT device having a first barrier layer underlying a second barrier layer.

FIG. 10 illustrates a flow diagram of some embodiments corresponding to the method in FIGS. 4-9.

DETAILED DESCRIPTION

The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numer-

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als and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

A high electron mobility transistor (HEMT) device includes a heterojunction which is at an interface between two materials having different band gaps and which acts as a channel region of the HEMT. For example, the heterojunction is disposed over a semiconductor substrate and can be disposed between an undoped layer (e.g., comprising a binary III-V semiconductor material such as gallium nitride) and a barrier layer (e.g., comprising a III-V semiconductor material such as aluminum gallium nitride). Additionally, a doped layer (e.g., comprising a binary III-V semiconductor material with a first doping type (e.g., p-type)) is arranged over the heterojunction. Further, a gate electrode is arranged over the doped layer and between a source contact and a drain contact to control current flow between the source and drain contacts of the HEMT.

During operation of the HEMT device, when a suitable voltage bias is applied across the gate electrode and the source and drain contacts, a current flows along the heterojunction. The applied voltage bias controls if the HEMT device operates in an enhancement mode or a depletion mode. In the enhancement mode, the HEMT device uses a gate to source voltage to switch the HEMT device “ON” (e.g., to “turn on” current between source and drain). Thus, in enhancement mode, the HEMT device is a “normally open” switch in some regards. In depletion mode, the HEMT device uses a gate to source voltage to switch the device “OFF” (e.g., to “turn off” current between source and drain). Thus, in depletion mode, the HEMT device is a “normally closed” switch in some regards.

For example, in the enhancement mode for n-channel HEMT devices, the conduction band at the heterojunction has a sharp valley that drops below and intersects the Fermi level, thereby forming a two-dimensional electron gas (2DEG) at the heterojunction when a gate to source voltage (e.g., a threshold voltage) is applied. A concentration of a first element (e.g., aluminum) within the barrier layer is relatively low (e.g., within a range of about 7-25%) due to a low activation rate of the dopants within the doped layer. For example, an activation rate of p-type dopants within a binary III-V semiconductor material (e.g., gallium nitride) is less than an activation rate of p-type dopants within a silicon substrate. By virtue of the relatively low concentration of the first element in the barrier layer, a concentration of 2DEG along the heterojunction is reduced, thereby ensuring the threshold voltage is positive. However, the decreased concentration of 2DEG along the heterojunction increases a resistance of the HEMT while in an “ON” state, and increases depletion of the 2DEG (e.g., resulting in a smaller breakdown voltage), thereby decreasing a performance of the HEMT device.

The present disclosure, in some embodiments, provides a method and a corresponding structure of an HEMT device

that comprises a first barrier layer and a second barrier layer with different concentrations of a first element (e.g., aluminum). For example, an undoped layer is formed over a substrate. A first barrier layer (e.g., comprising aluminum gallium nitride) is formed over the undoped layer, where the first element (e.g., aluminum) is a relatively low atomic percentage of the first barrier layer (e.g., within a range of about 7 to 25 atomic percent of the first barrier layer is the first element). A doped layer is formed over the first barrier layer. Subsequently, a second barrier layer (e.g., comprising aluminum gallium nitride) is selectively formed over the first barrier layer, where the first element (e.g., aluminum) is a relatively high atomic percentage of the second barrier layer (e.g., within a range of about 10 to 60 atomic percent of the second barrier layer is the first element). In some embodiments, an atomic percentage of the first element within the second barrier layer is greater than an atomic percentage of the first element within the first barrier layer. Further, the second barrier layer is formed in such a manner that it is laterally offset from a perimeter of the doped layer by a non-zero distance. Finally, a gate electrode is formed over the doped layer, a source contact is formed over the second barrier layer on a first side of the doped layer, and a drain contact is formed over the second barrier layer on a second side of the doped layer opposite the first side. Thus, a concentration of the first element (e.g., aluminum) underlying the doped layer is relatively low, thereby reducing a concentration of 2DEG around the doped layer and ensuring a positive threshold voltage. In addition, a concentration of the first element (e.g., aluminum) underlying the source and drain contacts is relatively high, thereby increasing a concentration of 2DEG around the source and drain contacts. This decreases resistance of the HEMT device while in an "ON" state and decreases depletion of the 2DEG, thereby increasing an endurance and breakdown voltage of the HEMT device. Therefore, an overall performance of the HEMT device is increased.

FIG. 1 illustrates a cross-sectional view of some embodiments of an integrated chip **100** having a first barrier layer **106** underlying a second barrier layer **108**.

The integrated chip **100** includes an undoped layer **104** overlying a substrate **102**. In some embodiments, the undoped layer **104** may, for example, be or comprise a binary III-V semiconductor material, such as, for example, aluminum nitride, gallium nitride, indium nitride, another suitable semiconductor material, or the like. Further, in some embodiments, the undoped layer **104** may be referred to as an undoped binary III-V semiconductor layer. In further embodiments, the substrate **102** may, for example, comprise silicon, monocrystalline silicon, silicon carbide, sapphire, a bulk silicon substrate, a silicon-on-insulator (SOI) substrate, or another suitable semiconductor substrate material.

A barrier structure **109** overlies the substrate **102**, where the barrier structure **109** comprises the first and second barrier layers **106**, **108**. The first barrier layer **106** overlies the undoped layer **104**. In some embodiments, the first barrier layer **106** may comprise a III-V semiconductor material, such as, for example, aluminum gallium nitride, indium aluminum gallium nitride, or another suitable III-V semiconductor material and/or has a first thickness t_1 . Further, the second barrier layer **108** overlies the first barrier layer **106**. In some embodiments, the second barrier layer **108** may comprise an III-V semiconductor material, such as, for example, aluminum gallium nitride, indium aluminum gallium nitride, or another suitable III-V semiconductor material and/or has a second thickness t_2 . In some embodiments, the first and second barrier layers **106**, **108** comprise

a same III-V semiconductor material (e.g., aluminum gallium nitride). In various embodiments, a concentration of a first element (e.g., aluminum) within the first barrier layer **106** is less than a concentration of the first element within the second barrier layer **108**. In some embodiments, the first element is aluminum, but other elements are amenable. In some embodiments, the first thickness t_1 is greater than the second thickness t_2 . In further embodiments, the first thickness t_1 is at least two times greater than the second thickness t_2 . In some embodiments, the first thickness t_1 is within a range of about 8 to 25 nanometers, a range of about 8 to 16 nanometers, or a range of about 16 to 25 nanometers. It will be appreciated that other values for the first thickness t_1 are also within the scope of the disclosure. In further embodiments, the second thickness t_2 is within a range of about 1 to 10 nanometers, a range of about 1 to 5 nanometers, or a range of about 5 to 10 nanometers. It will be appreciated that other values for the second thickness t_2 are also within the scope of the disclosure.

A high electrode mobility transistor (HEMT) device **112** is disposed over the first barrier layer **106**. The HEMT device **112** may include a doped layer **114** overlying the first barrier layer **106**. A first contact **118** overlies the second barrier layer **108** on a first side of doped layer **114**, and a second contact **120** overlies the second barrier layer **108** on a second side of the doped layer **114** that is opposite the first side. Thus, the first and second contacts **118**, **120** are separated from one another by the doped layer **114**. In addition, a gate electrode **116** overlies the doped layer **114** and is disposed within a dielectric structure **122**. In some embodiments, the doped layer **114** may comprise a same material (e.g., gallium nitride) as the undoped layer **104**. However, the doped layer **114** comprises a first doping type (e.g., p-type). In further embodiments, the doped layer **114** may, for example, be or comprise a binary III-V semiconductor material, such as, for example, aluminum nitride, gallium nitride, indium nitride, or another suitable semiconductor material with the first doping type. Further, in some embodiments, the doped layer **114** may be referred to as a doped binary III-V semiconductor layer. In addition, an atomic concentration of the first element within the barrier structure **109** may, for example, discretely increase along a first direction that extends from the middle region **103** to a peripheral region that laterally encloses the middle region **103**. The second barrier layer **108** is disposed within the peripheral region, and the first direction points from the doped layer **114** to the first contact **118** or the second contact **120**.

In some embodiments, a first atomic percentage of the first element (e.g., aluminum) within the first barrier layer **106** is within a range of about 7% to 25%, a range of about 7% to 16%, or a range of about 16% to 25%. It will be appreciated that other values for the first atomic percentage are also within the scope of the disclosure. Further, in some embodiments, a second atomic percentage of the first element (e.g., aluminum) within the second barrier layer **108** is within a range of about 10% to 60%, a range of about 10% to 35%, or a range of about 35% to 60%. It will be appreciated that other values for the second atomic percentage are also within the scope of the disclosure. In further embodiments, the first atomic percentage of the first element within the first barrier layer **106** is less than the second atomic percentage of the first element within the second barrier layer **108**. In yet further embodiments, the second barrier layer **108** has a different concentration of elements (e.g., a greater concentration of the first element) than the first barrier layer **106**. Further, the doped layer **114** and the

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gate electrode **116** are disposed laterally within a middle region **103**, such that the second barrier layer **108** abuts the middle region **103** and is laterally offset from a perimeter of the doped layer **114** by a non-zero distance.

In some embodiments, the HEMT device **112** is operated in an enhancement mode, where a threshold voltage (e.g., a gate to source voltage) is applied to the gate electrode **116** and the first contact **118** to switch the HEMT device **112** “ON” (e.g., to “turn on” current between the first contact **118** and the second contact **120**). Thus, in enhancement mode, the HEMT device **112** is a “normally open” switch in some regards (i.e., “normally open” in the absence of an applied voltage). Further, while applying the threshold voltage to the gate electrode **116** and the first contact **118**, a two-dimensional electron gas (2DEG) may form and/or flow along a heterojunction **111** at an interface between the undoped layer **104** and the first barrier layer **106**. In some embodiments, the heterojunction **111** may form due to a difference in band gap between the undoped layer **104** and the first barrier layer **106**. In addition, the 2DEG may form along the heterojunction **111** because, while applying the threshold voltage, the conductive band at the heterojunction **111** has a sharp valley that drops below and intersects the Fermi level. In some embodiments, in order to ensure the HEMT device **112** is a “normally open” switch, the p-type doping concentration of the doped layer **114** is sufficiently high enough to lift up the conduction band (e.g., above the Fermi level) in the middle region **103** of the HEMT device **112** while the HEMT device **112** is at equilibrium (i.e., in the absence of an applied threshold voltage). This is because the sufficiently high doping concentration of the p-type dopants ensures that there are a sufficient number of available holes to lift up the conduction band above the Fermi level while the HEMT device **112** is at equilibrium.

In some embodiments, dopant activation of p-type dopants (e.g., magnesium (Mg)) within a binary III-V semiconductor material (e.g., gallium nitride) is less and/or slower than dopant activation of p-type dopants within a silicon substrate. This may result in a decreased number of available holes in the doped layer **114**. Therefore, reducing the concentration of 2DEG along the heterojunction **111** within the middle region **103** (e.g., by the relatively low atomic percentage of aluminum in the first barrier layer **106**) reduces and/or mitigates a recombination of available holes from the doped layer **114** and electrons from the 2DEG along the heterojunction **111**. This ensures there are a sufficient number of available holes in the doped layer **114** to lift the conduction band above the Fermi level while the HEMT device **112** is at equilibrium, thereby ensuring the threshold voltage of the HEMT device **112** is positive. Further, in some embodiments, because the concentration of the first element (e.g., aluminum) within the second barrier layer **108** is greater than the concentration of the first element within the first barrier layer **106**, a concentration of 2DEG along the heterojunction **111** directly below the first and second contacts **118**, **120** is increased. This, in part, increases a number of charge carriers (e.g., 2DEG) near the first and second contacts **118**, **120**, thereby reducing a resistance of the HEMT device **112** while in an “ON” state and increasing a breakdown voltage of the HEMT device **112**. For example, the breakdown voltage of the HEMT device **112** may be greater than about 200 volts (V), within a range of about 200 to 1,100V, or another suitable voltage value. Therefore, the relatively low concentration of the first element in the first barrier layer **106** and the relatively high concentration of the first element in the second barrier layer **108** increases a performance and reliability of the HEMT device **112**. Fur-

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ther, in some embodiments, the first element (e.g., aluminum) may be referred to as a charge carrier enhancement element because it is configured to increase a concentration of the charge carriers (e.g., 2DEG) disposed along the heterojunction **111**.

In some embodiments, if the first atomic percentage of the first element (e.g., aluminum) within the first barrier layer **106** is substantially low (e.g., less than about 7%), then a reliability of the HEMT device **112** is reduced due to electron surface trapping within the middle region **103**. In further embodiments, if the first atomic percentage of the first element within the first barrier layer **106** is substantially large (e.g., greater than about 25%), then the threshold voltage of the HEMT device **112** may be negative due to an increased concentration of 2DEG within the middle region **103** while the HEMT device **112** is at equilibrium, such that the HEMT device **112** may not be a “normally open” switch. In yet further embodiments, if the second atomic percentage of the first element (e.g., aluminum) within the second barrier layer **108** is substantially small (e.g., less than about 10%), then a concentration of 2DEG along the heterojunction **111** and directly beneath the first and second contacts **118**, **120** is substantially low, thereby increasing a resistance of the HEMT device **112** while in an “ON” state. In various embodiments, if the second atomic percentage of the first element within the second barrier layer **108** is substantially large (e.g., greater than about 60%), then a concentration of 2DEG is increased, thereby increasing an electric field in the HEMT device **112** while in an “ON” state and decreasing a breakdown voltage of the HEMT device **112**.

In some embodiments, if the first thickness t_1 is relatively low (e.g., less than about 8 nanometers), then a reliability of the HEMT device **112** is reduced due to electron surface trapping. In further embodiments, if the first thickness t_1 is relatively large (e.g., greater than about 25 nanometers), then the threshold voltage of the HEMT device **112** may be negative due to an increased concentration of 2DEG within the middle region **103** while the HEMT device **112** is at equilibrium, such that the HEMT device **112** may not be a “normally open” switch. In yet further embodiments, if the second thickness t_2 is relatively low (e.g., less than about 1 nanometer), then a reliability of the HEMT device **112** is reduced due to electron surface trapping. In yet further embodiments, if the second thickness t_2 is relatively high (e.g., greater than about 10 nanometers), then a concentration of 2DEG is increased, thereby increasing an electric field in the HEMT device **112** while in an “ON” state and decreasing a breakdown voltage of the HEMT device **112**. Thus, in some embodiments, a ratio between the first thickness t_1 and the second thickness t_2 (i.e., $t_1:t_2$) may be about 8:1, 2.5:1, or within a range of about 1:1 to 10:1. It will be appreciated that other values for the ratio between the first thickness t_1 and the second thickness t_2 are also within the scope of the disclosure.

FIG. 2 illustrates a cross-sectional view of an integrated chip **200** according to some alternative embodiments of the integrated chip **100** of FIG. 1, in which a bottom surface **108bs** of the second barrier layer **108** is disposed below a top surface **106ts** of the first barrier layer **106** by a distance dl . In such embodiments, the first barrier layer **106** comprises the first thickness t_1 within the middle region **103** and the second barrier layer **108** comprises the second thickness t_2 within a peripheral region laterally outside the middle region **103**. Further, the first barrier layer **106** may, for example, comprise a third thickness t_3 within the peripheral region that is less than the first thickness t_1 . In some embodiments, the third thickness t_3 is greater than the second thickness t_2 .

In various embodiments, the distance d_1 may be within a range of about 0.5 to 1.5 nanometers, about 1.5 to 3 nanometers, about 0.5 to 3 nanometers, or another suitable value. Further, the distance d_1 may, for example, be due to the second barrier layer **108** being formed by a selective growth process with a relatively high growth temperature (e.g., within a range of about 700 to 1,200 Celsius), such that at least a portion of the first barrier layer **106** decomposes and/or is removed during the selective growth process. This results in the reduced third thickness t_3 of the second barrier layer **108** in the peripheral region that is laterally offset from the middle region **103**.

FIG. 3 illustrates a cross-sectional of an integrated chip **300** according to some alternative embodiments of the integrated chip **100** of FIG. 1.

A lower buffer layer **302** is disposed between the substrate **102** and the undoped layer **104**. In some embodiments, the lower buffer layer **302** comprises a III-V semiconductor material, such as, for example, aluminum nitride, indium nitride, gallium nitride, aluminum gallium nitride, another suitable material, or any combination of the foregoing. In further embodiments, the lower buffer layer **302** may comprise a single layer comprising the III-V semiconductor material or a multi-layer stack each comprising the III-V semiconductor material (each layer in the multi-layer stack may comprise a different III-V semiconductor material). The lower buffer layer **302** may, for example, serve to compensate for a difference in lattice constants, crystalline structures, thermal expansion coefficients, or any combination of the foregoing between the substrate **102** and overlying layers (e.g., the undoped layer **104**, the first barrier layer **106**, the second barrier layer **108**, etc.). Further, a passivation layer **304** is disposed between the dielectric structure **122** and the first barrier layer **106**. In some embodiments, the passivation layer **304** may, for example, be or comprise silicon dioxide, silicon nitride, silicon oxynitride, silicon oxycarbide, another suitable dielectric material, or any combination of the foregoing.

The integrated chip **300** includes conductive vias **306** embedded within the dielectric structure **122**. In some embodiments, the dielectric structure **122** may, for example, be or comprise silicon dioxide, a low-k dielectric material, an extreme low-k dielectric material, another suitable dielectric material, or any combination of the foregoing. In further embodiments, the conductive vias **306** may, for example, be or comprise titanium nitride, tantalum nitride, copper, tungsten, aluminum, ruthenium, another suitable conductive material, or any combination of the foregoing. The first contact **118**, the second contact **120**, and the gate electrode **116** may be electrically coupled to one of the conductive vias **306**. The conductive vias **306** may be coupled to voltage sources to control the operation (e.g., "ON" and/or "OFF") of the HEMT device **112**. In some embodiments, up to 20 volts may be applied across the HEMT device **112**. In other embodiments, the voltage bias applied to the HEMT device **112** may be within a range of about 1 volt to 12 volts. It will be appreciated that other values for the voltage bias applied to the HEMT device **112** are also within the scope of the disclosure. In further embodiments, the gate electrode **116** may be vertically offset from the doped layer **114** (not shown), such that the dielectric structure **122** is disposed between a bottom surface of the gate electrode **116** and a top surface of the doped layer **114**.

FIGS. 4-9 illustrate cross-sectional views **400-900** of some embodiments of a method for forming an integrated chip having an HEMT device with a first barrier layer underlying a second barrier layer according to the present

disclosure. Although the cross-sectional views **400-900** shown in FIGS. 4-9 are described with reference to a method, it will be appreciated that the structures shown in FIGS. 4-9 are not limited to the method but rather may stand alone separate of the method. Although FIGS. 4-9 are described as a series of acts, it will be appreciated that these acts are not limited in that the order of the acts can be altered in other embodiments, and the methods disclosed are also applicable to other structures. In other embodiments, some acts that are illustrated and/or described may be omitted in whole or in part.

As shown in cross-sectional view **400** of FIG. 4, a substrate **102** is provided and a lower buffer layer **302** is deposited over the substrate **102**. In some embodiments, the substrate **102** comprises silicon, sapphire, or silicon carbide. In some embodiments, the lower buffer layer **302** comprises a III-V semiconductor material, such as, for example, aluminum nitride, indium nitride, gallium nitride, aluminum gallium nitride, another suitable material, or any combination of the foregoing. Further, an undoped layer **104** is deposited over the lower buffer layer **302**. In further embodiments, the undoped layer **104** may, for example, be or comprise a III-V semiconductor material, or a III-V semiconductor material, such as, for example, aluminum nitride, indium nitride, gallium nitride, or the like.

Also illustrated in FIG. 4, a first barrier layer **106** is deposited over the undoped layer **104** with a first thickness t_1 . In some embodiments, the first barrier layer **106** may, for example, be or comprise a III-V semiconductor material, such as, for example, aluminum gallium nitride, or another suitable III-V semiconductor material. In some embodiments, the first barrier layer **106** is formed in a manner, such that a first atomic percentage of a first element (e.g., aluminum) within the first barrier layer **106** is, for example, within a range of about 7% to 25%, a range of about 7% to 16%, or a range of about 16% to 25%. It will be appreciated that other values for the first atomic percentage are also within the scope of the disclosure. In some embodiments, the first thickness t_1 is within a range of about 8 to 25 nanometers, a range of about 8 to 16 nanometers, or a range of about 16 to 25 nanometers. It will be appreciated that other values for the first thickness t_1 are also within the scope of the disclosure.

Further, a doped layer **114** is formed over the first barrier layer **106**. In some embodiments, the doped layer **114** may comprise a same material as the undoped layer **104**. However, the doped layer **114** comprises a first doping type (e.g. p-type). In further embodiments, the doped layer **114** may, for example, be or comprise a III-V semiconductor material, such as, for example, aluminum nitride, gallium nitride, indium nitride, another suitable III-V semiconductor material, or the like with the first doping type. In some embodiments, a process for forming the doped layer **114** may include: depositing a III-V semiconductor layer over the first barrier layer **106**; forming a masking layer (not shown) over the III-V semiconductor layer; patterning the III-V semiconductor layer according the masking layer, thereby defining the doped layer **114**; and performing a removal process to remove the masking layer. In further embodiments, after depositing the III-V semiconductor layer, a selective ion implantation process may be performed on the III-V semiconductor layer, where dopants (e.g., magnesium (Mg)) are selectively implanted in the III-V semiconductor layer such that the III-V semiconductor layer comprises the first doping type (e.g., p-type). In yet further embodiments, a deposition

process utilized to form the doped layer **114** may include in-situ doping the III-V semiconductor layer with p-type dopants.

In some embodiments, the lower buffer layer **302**, the undoped layer **104**, the first barrier layer **106**, and/or the doped layer **114** may be deposited by an epitaxial process (e.g., vapor-phase epitaxy (VPE), liquid-phase epitaxy (LPE), molecular-beam epitaxy (MBE), some other epitaxial process, or a combination of the foregoing) and/or another form of a deposition process (e.g., chemical vapor deposition (CVD), metal organic chemical vapor deposition (MO-CVD), plasma enhanced chemical vapor deposition (PE-CVD), atomic layer deposition (ALD), physical vapor deposition (PVD), sputtering, electron beam/thermal evaporation, any combination of the foregoing, etc.). For example, the first barrier layer **106** may be formed by VPE, LPE, MBE, some other epitaxial process, or a combination of the foregoing. In another example, the doped layer **114** may be formed by an epitaxial process (e.g., VPE, LPE, MBE, some other epitaxial process, or a combination of the foregoing) that in-situ dopes the doped layer **114** with the first doping type (e.g., p-type).

As shown in cross-sectional view **500** of FIG. **5**, a masking layer **502** is formed over the doped layer **114**. The masking layer **502** and the undoped layer **104** are spaced laterally within a middle region **103**. In some embodiments, the masking layer **502** may, for example, be or comprise a photosensitive material, a hard masking material, silicon nitride, silicon carbide, another suitable masking material, or any combination of the foregoing. The masking layer **502** may, for example, be or comprise an oxide, such as silicon dioxide, or another suitable material.

As shown in cross-sectional view **600** of FIG. **6**, a selective growth process is performed to selectively deposit a second barrier layer **108** over the first barrier layer **106** in regions laterally offset from the masking layer **502**, thereby forming a barrier structure **109**. The barrier structure **109** includes the first barrier layer **106** and the second barrier layer **108**. In some embodiments, the selective growth process is performed in such a manner to ensure the second barrier layer **108** is not formed over the masking layer **502**. In some embodiments, the second barrier layer **108** may, for example, be or comprise a III-V semiconductor material, such as, for example, aluminum gallium nitride. In further embodiments, an atomic percentage of the first element (e.g., aluminum) within the second barrier layer **108** is within a range of about 10% to 60%, a range of about 10% to 35%, or a range of about 35% to 60%. It will be appreciated that other values for the second atomic percentage are also within the scope of the disclosure. In further embodiments, the second barrier layer **108** is formed to a second thickness **t2** that is within a range of about 1 to 10 nanometers, a range of about 1 to 5 nanometers, or a range of about 5 to 10 nanometers. It will be appreciated that other values for the second thickness **t2** are also within the scope of the disclosure. The selective growth process may be an epitaxial process or another form of a deposition process (e.g., CVD, MO-CVD, PE-CVD, ALD, PVD, sputtering, electron beam/thermal evaporation, etc.) performed at a relatively high regrowth temperature. For example, in some embodiments, the selective growth process may be performed by MO-CVD. Thus, the first barrier layer **106** may be formed by a first deposition process (e.g., an epitaxial process) while the second barrier layer **108** is formed by a second deposition process (e.g., a deposition process such as MO-CVD) different from the first deposition process. In some embodiments, the relatively high regrowth tempera-

ture may be within a range of about 700 to 1,200 Celsius, or some other suitable temperature value.

In further embodiments, due to the second deposition process and/or the relatively high regrowth temperature used to form the second barrier layer **108**, at least a portion of the first barrier layer **106** laterally offset from the masking layer **502** decomposes and/or is removed. This results in a bottom surface **108bs** of the second barrier layer **108** being disposed below a top surface **106ts** of the first barrier layer **106** by a distance **dl**. In such embodiments, the first barrier layer **106** comprises the first thickness **t1** within the middle region **103** and the first barrier layer **106** comprises a third thickness **t3** within a region laterally offset from the middle region **103**. The first thickness **t1** is greater than or equal to the third thickness **t3**. In some embodiments, the third thickness **t3** is greater than the second thickness **t2**. In various embodiments, the distance **dl** may be within a range of about 0.5 to 1.5 nanometers, about 1.5 to 3 nanometers, about 0.5 to 3 nanometers, or another suitable value. Further, the third thickness **t3** may be within a range of about 7.5 to 24.5 nanometers, a range of about 5 to 22 nanometers, or another suitable thickness value.

As shown in cross-sectional view **700** of FIG. **7**, a removal process is performed to remove the masking layer (**502** of FIG. **6**). In some embodiments, the removal process includes performing a wet etch process.

As shown in cross-sectional view **800** of FIG. **8**, passivation layer **304** is formed over the first barrier layer **106**. In some embodiments, the passivation layer **304** may, for example, be deposited by CVD, PVD, ALD, or another suitable deposition or growth process. Subsequently, a gate electrode **116** is formed over the doped layer **114**, and a first contact **118** and a second contact **120** are formed over the second barrier layer **108**, thereby forming an HEMT device **112**. In some embodiments, a process for forming the gate electrode **116**, the first contact **118**, and the second contact **120** may include: patterning the passivation layer **304** to define a plurality of openings (not shown); depositing (e.g., by CVD, PVD, sputtering, electroless plating, electroplating, or another suitable deposition or growth process) a conductive layer (e.g., comprising titanium nitride, nickel, tungsten, titanium, platinum, titanium, aluminum, copper, tungsten, another suitable conductive material, or any combination of the foregoing) over the passivation layer **304**, such that the conductive layer fills the plurality of openings; forming a masking layer (not shown) over the conductive layer; patterning the conductive layer according to the masking layer, thereby defining the gate electrode **116**, and the first and second contacts **118**, **120**; and performing a removal process to remove the masking layer.

As shown in the cross-sectional view **900** of FIG. **9**, a dielectric structure **122** is formed over the HEMT device **112**. Subsequently, a plurality of conductive vias **306** are formed within the dielectric structure **122**. In some embodiments, the dielectric structure **122** may, for example, be deposited by CVD, PVD, ALD, or another suitable growth or deposition process. In further embodiments, the dielectric structure **122** may, for example, be or comprise silicon dioxide, a low-k dielectric material, an extreme low-k dielectric material, another suitable dielectric material, or any combination of the foregoing. In further embodiments, the conductive vias **306** may be formed by a single damascene process. In addition, in various embodiments, the conductive vias **306** may, for example, be or comprise copper, aluminum, tungsten, titanium nitride, tantalum nitride, ruthenium, another suitable conductive material, or any combination of the foregoing.

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FIG. 10 illustrates a flow diagram of some embodiments of a method 1000 of forming an integrated chip with an HEMT device having a first barrier layer underlying a second barrier layer.

While the method 1000 is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

At act 1002, an undoped layer is deposited over a substrate.

At act 1004, a first barrier layer is deposited over the undoped layer, such that the first barrier layer comprises a III-V semiconductor material with a first atomic percentage of a first element. The first barrier layer is deposited by a first deposition process.

At act 1006, a doped layer is formed over the first barrier layer, where the doped layer is disposed within a middle region of the substrate. FIG. 4 illustrates a cross-sectional view 400 of some embodiments corresponding to acts 1002, 1004, and 1006.

At act 1008, a masking layer is formed over the doped layer, where the masking layer is disposed within the middle region. FIG. 5 illustrates a cross-sectional view 500 of some embodiments corresponding to act 1008.

At act 1010, a second barrier layer is selectively grown over the first barrier layer, such that the second barrier layer comprises the III-V semiconductor material with a second atomic percentage of the first element. The second atomic percentage is greater than the first atomic percentage. The second barrier layer is laterally offset from the middle region, and the selective growth process is different from the first deposition process. FIG. 6 illustrates a cross-sectional view 600 of some embodiments corresponding to act 1010.

At act 1012, first and second contacts are formed over the second barrier layer.

At act 1014, a gate electrode is formed on the doped layer, such that the gate electrode is spaced laterally between the first and second contacts. FIG. 8 illustrates a cross-sectional view 800 of some embodiments corresponding to acts 1012 and 1014.

Accordingly, in some embodiments, the present disclosure relates to a high electron mobility transistor (HEMT) device having a first barrier layer underlying a second barrier layer. The first barrier layer comprises a III-V semiconductor material with a first atomic percentage of a first element, and the second barrier layer comprises the III-V semiconductor material with a second atomic percentage of the first element, where the second atomic percentage is greater than the first atomic percentage.

In some embodiments, the present application provides an integrated chip, including: an undoped layer overlying a substrate; a first barrier layer overlying the undoped layer; a doped layer overlying the first barrier layer; and a second barrier layer overlying the first barrier layer, wherein the second barrier layer is laterally offset from a perimeter of the doped layer by a non-zero distance, wherein the first and second barrier layers comprise a same III-V semiconductor material, wherein a first atomic percentage of a first element

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within the first barrier layer is less than a second atomic percentage of the first element within the second barrier layer.

In some embodiments, the present application provides an integrated chip, including: a buffer layer overlying a substrate; an undoped binary III-V semiconductor layer overlying the buffer layer; a doped binary III-V semiconductor layer overlying the undoped binary III-V semiconductor layer, wherein the doped binary III-V semiconductor layer is disposed within a middle region of the substrate; and a barrier structure overlying the undoped binary III-V semiconductor layer, wherein the barrier structure comprises a III-V semiconductor material with a first element, wherein an atomic concentration of the first element discretely increases along a first direction that extends from the middle region to a peripheral region, where the peripheral region laterally surrounds the middle region, and wherein the first direction points away from the doped binary III-V semiconductor layer.

In some embodiments, the present application provides a method for forming an integrated chip, the method includes: depositing an undoped layer over a substrate; depositing a first barrier layer over the undoped layer, wherein the first barrier layer comprises a III-V semiconductor material with a first atomic percentage of a first element; forming a doped layer over the first barrier layer; selectively forming a second barrier layer over the first barrier layer such that the second barrier layer is laterally offset from a perimeter of the doped layer by a non-zero distance, wherein the second barrier layer comprises the III-V semiconductor material with a second atomic percentage of the first element, wherein the second atomic percentage is greater than the first atomic percentage; and forming a gate electrode over the doped layer.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An integrated chip, comprising:

- an undoped layer overlying a substrate;
- a first barrier layer overlying the undoped layer;
- a doped layer overlying the first barrier layer; and
- a second barrier layer overlying the first barrier layer, wherein the second barrier layer is laterally offset from a perimeter of the doped layer by a non-zero distance, wherein the first and second barrier layers comprise a same III-V semiconductor material, wherein a first atomic percentage of a first element within the first barrier layer is less than a second atomic percentage of the first element within the second barrier layer, wherein the first barrier layer comprises a first segment having a first thickness below the doped layer and a second segment having a second thickness directly adjacent to the first segment, wherein the second barrier layer has a third thickness, wherein the first thickness is greater than the second thickness and the third thickness, wherein the first segment has a top surface

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contacting a bottom surface of the doped layer and the second segment has a top surface contacting a bottom surface of the second barrier layer, wherein a bottom surface of the first segment is aligned with a bottom surface of the second segment, wherein the first segment contacts an inner sidewall of the second barrier layer.

2. The integrated chip according to claim 1, wherein the first element is aluminum.

3. The integrated chip according to claim 1, wherein the same III-V semiconductor material is aluminum gallium nitride.

4. The integrated chip according to claim 1, wherein the first thickness of the first segment is defined between a bottom surface of the first barrier layer and the bottom surface of the doped layer, wherein the second thickness of the second segment is defined between the bottom surface of the first barrier layer and the bottom surface of the second barrier layer.

5. The integrated chip according to claim 1, wherein the bottom surface of the second barrier layer is disposed below a top surface of the first barrier layer by a first distance, wherein a height of the second barrier layer above the top surface of the first barrier layer is less than the first distance.

6. The integrated chip according to claim 1, wherein the second thickness is greater than the third thickness.

7. The integrated chip according to claim 1, further comprising:

- a gate electrode overlying the doped layer;
- a first contact overlying the second barrier layer; and
- a second contact overlying the second barrier layer, wherein the first and second contacts are separated from one another by the doped layer.

8. The integrated chip according to claim 1, further comprising:

- a passivation layer overlying the first barrier layer, wherein the passivation layer comprises a U-shaped segment arranged between the doped layer and the inner sidewall of the second barrier layer.

9. An integrated chip, comprising:

- a buffer layer overlying a substrate;
- an undoped binary III-V semiconductor layer overlying the buffer layer;

- a doped binary III-V semiconductor layer overlying the undoped binary III-V semiconductor layer, wherein the doped binary III-V semiconductor layer is disposed within a middle region of the substrate and comprises a first outer sidewall opposite a second outer sidewall; and

- a barrier structure overlying the undoped binary III-V semiconductor layer, wherein the barrier structure comprises a III-V semiconductor material with a first element, wherein an atomic concentration of the first element discretely increases at a first point along a first direction that extends from the middle region to a peripheral region adjacent to the middle region, wherein the first direction points away from the doped binary III-V semiconductor layer, wherein a lateral distance between the first point and the first outer sidewall of the doped binary III-V semiconductor layer facing the first direction is greater than a distance between the first and second outer sidewalls of the doped binary III-V semiconductor layer, wherein the barrier structure comprises:

- a first barrier layer disposed between the undoped binary III-V semiconductor layer and the doped

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binary III-V semiconductor layer, wherein the first barrier layer comprises the III-V semiconductor material; and

- a second barrier layer overlying the first barrier layer, wherein the second barrier layer is laterally offset from the middle region, wherein the second barrier layer comprises the III-V semiconductor material with a higher atomic concentration of the first element relative to the first barrier layer, wherein the first barrier layer contacts opposing inner sidewalls of the second barrier layer that are offset from the first and second outer sidewalls of the doped binary III-V semiconductor layer.

10. The integrated chip according to claim 9, wherein a thickness of the barrier structure discretely increases along the first direction.

11. The integrated chip according to claim 9, wherein the first element is aluminum.

12. The integrated chip according to claim 9, further comprising:

- a first contact overlying the barrier structure and within the peripheral region, wherein the first contact is on a first side of the doped binary III-V semiconductor layer; and

- a second contact overlying the barrier structure and within the peripheral region, wherein the second contact is on a second side of the doped binary III-V semiconductor layer, wherein the first side is opposite the second side.

13. The integrated chip according to claim 12, wherein the atomic concentration of the first element directly underlying the doped binary III-V semiconductor layer is less than the atomic concentration of the first element directly underlying the first and second contacts.

14. The integrated chip according to claim 12, wherein bottom surfaces of the first and second contacts are disposed above a bottom surface of the doped binary III-V semiconductor layer.

15. The integrated chip according to claim 14, wherein the bottom surfaces of the first and second contacts are disposed below a top surface of the doped binary III-V semiconductor layer.

16. An integrated chip, comprising:

- an undoped layer overlying a substrate;
- a first barrier layer overlying and directly contacting the undoped layer, wherein the first barrier layer comprises a first III-V semiconductor material having a first concentration of elements;

- a doped layer overlying and directly contacting the first barrier layer; and

- a second barrier layer overlying and directly contacting the first barrier layer, wherein the second barrier layer comprises the first III-V semiconductor material having a second concentration of elements different from the first concentration of elements, wherein the doped layer is spaced laterally between opposing sidewalls of the second barrier layer, wherein a first thickness of the first barrier layer is greater than a second thickness of the second barrier layer, wherein the second barrier layer directly contacts a first sidewall of the first barrier layer, wherein the first sidewall of the first barrier layer is laterally offset from the doped layer by a lateral distance.

17. The integrated chip of claim 16, wherein the first barrier layer comprises a first atomic percentage of aluminum and the second barrier layer comprises a second atomic percentage of aluminum less than the first atomic percentage, wherein the first atomic percentage is within a range of

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about 7 to 25 percent and the second atomic percentage is within a range of about 10 to 60 percent.

18. The integrated chip of claim **16**, further comprising:
a first contact overlying and directly contacting the second barrier layer; and

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a second contact overlying and directly contacting the second barrier layer, wherein the first and second contacts are arranged on opposing sides of the doped layer.

19. The integrated chip of claim **16**, wherein the doped layer comprises a second III-V semiconductor material different from the first III-V semiconductor material, wherein a third thickness of the doped layer is greater than the second thickness.

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20. The integrated chip of claim **16**, wherein the lateral distance is greater than a width of the doped layer.

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